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Key Points:

- We developed AI-PAL, a catalog construction workflow that trains an artificial intelligence (AI) phase picker with PAL detections, an established rule-based algorithm
- AI-PAL demonstrated greater temporal stability than matched filter, while also offering improved scalability compared to pretrained models
- The performance of AI picker is influenced by the quality of training data, with different models adapting to various sampling strategies

Supporting Information:

Supporting Information may be found in the online version of this article.

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AI-PAL: Self-Supervised AI Phase Picking via Rule-Based Algorithm for Generalized Earthquake Detection

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Abstract Delineating fault structures through microseismicity is crucial for earthquake hazard assessment, yet constructing high-resolution catalogs over extended periods remains challenging. This study introduces AI-PAL, a novel deep learning-driven workflow that employs a Self-Attention RNN (SAR) model trained with detections from PAL, an established rule-based algorithm (Zhou, Yue, et al., 2021, <https://doi.org/10.1785/0220210111>), for generalized earthquake detection. PAL utilizes short-term-average over long-term-average algorithm for event detection, ensuring consistent performance across different datasets. AI-PAL leverages these rule-based picks as training labels, enabling self-supervised learning of the SAR model across arbitrary regions, thereby enhancing PAL's detection capabilities. We applied SAR-PAL to two distinct regions that are featured by recent large earthquakes: (a) the preseismic period of the Ridgecrest-Coso region (2008–2019), and (b) the pre-to-postseismic period of the East Anatolian Fault Zone (EAFZ, 2020–2023/04). Our results demonstrate that SAR-PAL offers slightly higher detection completeness than the quake template matching matched filter catalog, while boosts over 100 times faster processing and a superior temporal stability, avoiding detection gaps during background periods. Compared to PhaseNet and GAMMA, two widely recognized phase picker and associator, SAR-PAL proved more scalable, achieving ~2.5 times more event detections in the EAFZ case, along with a ~7 times higher phase association rate. We further experimented training PhaseNet and SAR with PAL detections and routine catalogs, and found that no other combinations matched the detection performance of SAR-PAL. The enhanced catalogs built by SAR-PAL reveals geometrical complexities of the Ridgecrest faults and the Erkenek-Pütürge segment of EAFZ, offering insights into their contrasting roles during the large earthquake.

Plain Language Summary Seismic monitoring helps scientists understand fault structures and assess earthquake hazards, but building detailed earthquake catalogs over long periods remains difficult. Our study introduces AI-PAL, a new deep learning approach that improves earthquake detection by combining artificial intelligence with rule-based methods. AI-PAL uses an existing algorithm, PAL, to automatically generate training labels, allowing a neural network to learn from regional seismic data without requiring manually labeled events. This self-supervised learning approach enables more accurate and efficient detection of small earthquakes. We tested AI-PAL in two earthquake-prone regions: the Ridgecrest-Coso area in California and the East Anatolian Fault Zone in Turkey. Our results show that AI-PAL detects more earthquakes than traditional methods while being 100 times faster and more reliable over time. Compared to other AI-based seismic tools, AI-PAL identified 2.5 times more earthquakes and associated seismic phases with 7 times greater accuracy in Turkey. These improvements allow scientists to build more complete earthquake catalogs, revealing fault structures that were previously undetectable. By applying AI-PAL, researchers can better monitor earthquake activity and improve hazard assessments, ultimately supporting safer communities in seismically active regions.

1. Introduction

Microseismicity provides a direct indication to the fault structure and slip behavior at depth. Such strategy is especially useful when the fault slips at a high rate, for example, during early aftershock period (e.g., Ding et al., 2023; Shelly et al., 2024; Zhou, Ren, et al., 2022), or other intense seismic sequences (e.g., Gong & Fan, 2022; Ross et al., 2020; Tan et al., 2021). However, a huge portion of faults generate low seismicity rate during the interseismic period because of a high locking ratio or a low fault slip rate (e.g., Bletery et al., 2020; Chamberlain et al., 2021; Jiang & Lapusta, 2016; Uchida & Bürgmann, 2021; Zhou, Yue, et al., 2022), while,

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unfortunately, these faults are also prone to large earthquakes (Lay & Nishenko, 2022; Sykes, 2021). To study such low-seismicity faults, a long-term observation is always necessary. For example, Schurr et al. (2020) built a seismic catalog for 7 years before the 2014 Iquique earthquake, and found that the pre-mainshock seismicity complements the coseismic slip; Sukan et al. (2023) observed a 8 year migration of seismicity toward the nucleation area of the 2016 central Italy seismic sequence. Technically, the construction of long-term catalogs requires a workflow that is both computationally efficient and of high detection completeness, which is still a challenging task.

Currently, two types of cataloging workflow can realize a state-of-the-art performance: (a) the PAL style workflow that follows “phase Picking - phase Association - event Location” procedure (e.g., Zhang et al., 2022; Zhou, Yue, et al., 2021; Zhu, Hou, et al., 2022), and (b) the matched filter technique (MFT) that utilizes pre-detected events as templates to detect similar events (e.g., Neves et al., 2022; Ross, Idini, et al., 2019; Shelly, 2020). The detection completeness of PAL style workflows is basically dependent on the phase picking algorithm. In recent years, algorithms based on artificial intelligence (AI), specifically deep learning, realize outstanding phase picking performance in terms of the detectability and picking precision (e.g., Mousavi et al., 2020; Sun et al., 2023; Yu & Wang, 2022; Zhou et al., 2019; Zhu & Beroza, 2018). Most of these models, for example, PhaseNet (Zhu & Beroza, 2018), are trained on regional or global datasets, aiming at building a generalized model that works for various data not included in the training set. However, systematic tests show that the AI pickers can suffer from inconsistent performance among data in different regions (e.g., Chai et al., 2020; Jiang et al., 2021; Münchmeyer et al., 2022; Park et al., 2023; Zhu, Li, & Fang, 2022), especially for ocean-bottom seismograms (Bornstein et al., 2024; Niksejel & Zhang, 2024; Xi et al., 2024) and distributed acoustic sensing data (Zhu et al., 2023). This indicates a lower generalizability compared with traditional rule-based algorithms, such as short-term-average over long-term average (STA/LTA). On the other hand, the MFT methods can realize even higher detection ability than AI pickers (e.g., Mousavi et al., 2019; Yoon & Shelly, 2024; Zhou, Ghosh, et al., 2021), but its low computational efficiency makes it difficult to process big data (Ross, Trugman, et al., 2019). Moreover, the detection results of MFT may be biased by incomplete templates (e.g., Herrmann & Marzocchi, 2020). Thus, the AI-based picker is the most promising method that combines both high efficiency and high detectability, whereas further improvements are needed to realize a consistent picking performance on a large spatiotemporal range of data.

In this paper, we introduce AI-PAL, a novel cataloging workflow featured by self-supervised learning of AI picker enabled by rule-based algorithm. This approach effectively addresses the generalization problem faced by deep learning models. We apply the AI-PAL workflow on two cases that covers a local-to-regional scale and years-to-decade time length, in order to demonstrate its advantage in building long-term catalogs.

2. Methods

2.1. Overview of the AI-PAL Workflow

As reviewed in the Introduction section, the first-generation AI pickers are commonly used as pre-trained models across various datasets, but have been shown to lack robust generalization. To address this limitation, we designed a new workflow that automatically generates a localized training set to obtain a locally optimized neural network (Figure 1). This process follows the principles of self-supervised learning, where the model is trained using labels generated by a rule-based algorithm rather than relying on manually labeled data. The AI-PAL workflow is composed of a detection module followed by a location module, and the detection module utilized this self-supervised learning idea.

2.1.1. Detection Modules

In the model training step of the earthquake detection module, we utilize the PAL method (Zhou, Yue, et al., 2021) to construct a local training set (Figure 1). PAL utilizes rule-base algorithms for phase picking and association, thus does not have generalization problem as for deep learning. Tests on the 2019 Ridgecrest aftershock sequence show that PAL realizes around two times the number of detections by the Southern California Seismic Network (SCSN) (Zhou, Yue, et al., 2021), and PAL has been successfully applied in multiple regions and seismic sequences (e.g., Ding et al., 2023; Zhou, Ghosh, et al., 2021; Zhou, Yue, et al., 2022). Since the detection of PAL is basically made by the STA/LTA algorithm, which detects both earthquakes and pulse-like noises (e.g., anthropogenic noise or data glitches), we usually set a relatively high triggering threshold to avoid

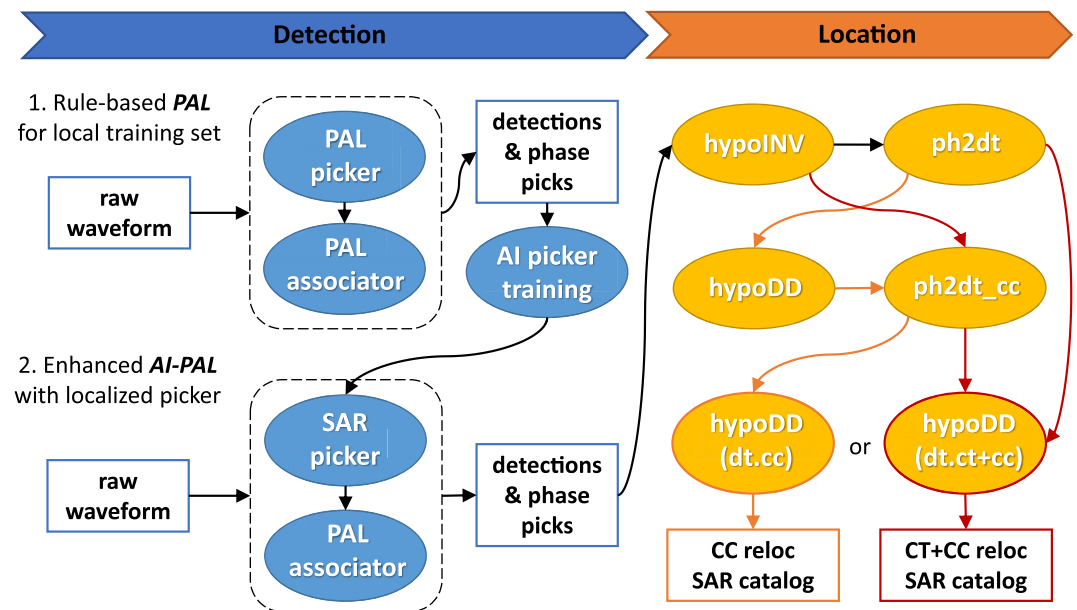


Figure 1. The AI-PAL workflow. The blue and yellow modules denote the detection and location algorithms, respectively. PAL refers to a rule-based cataloging workflow developed by Zhou, Yue, et al. (2021); Self-Attention RNN (SAR) refers to the SAR phase picker model developed in this study. CT/dt.ct and CC/dt.cc refer to catalog-based and cross-correlation-based differential travel time, following the hypoDD terminology. The ph2dt_cc module is a CC-based differential time calculation method developed in this study.

high false detection ratio and to reliably detect high signal-to-noise ratio (SNR) events. The PAL detections serve as the training set for our deep learning model, which can realize stable detections for weak signals. Note that neural networks usually require a large number of training samples to tune the model weights, but the number of training samples is dependent on the model complexity. Thus, for light-weight models (e.g., Yu & Wang, 2022; Zhou et al., 2019), as that used in this study (see the next subsection for details), a relatively small training set is required to optimize the neural network (Mousavi et al., 2020; Zhou, Ghosh, et al., 2021).

After the model training step, we simply substitute the PAL picker with the locally trained AI picker and associate with PAL associator (Figure 1). The PAL associator groups pairs of *P* & *S* picks into events based on their travel time-location relationship, and it also obtains a grid-searched location in the meantime. The magnitude calculation is also completed by the PAL associator in the local magnitude scale (M_L), based on the *S*-wave amplitude and hypocentral distance (refer to Zhou, Yue, et al., 2021 for more details). We will show in this paper that this AI-PAL workflow realizes >2.5 times more event detections compared to PAL, and is of a much higher detection stability and accuracy. It also generalizes well among very different tectonic settings, spatial scales, and network configurations.

2.1.2. Location Modules

To locate and relocate the detections, we build interfaces for HypoInverse (abbreviated as HypoINV thereafter, Klein, 2002) and HypoDD (Waldhauser, 2001; Waldhauser & Ellsworth, 2000) (Figure 1). HypoINV is a widely adopted algorithm for absolute location, which minimizes the travel time residuals in an iterative manner, where the weight for the phases are adaptively determined based on their epicentral distance and residual time (Klein, 2002). HypoDD is a double-difference (DD) relocation algorithm that minimizes the differential travel time residuals between event pairs to constrain their relative locations (Waldhauser & Ellsworth, 2000). Given that the fault structure is manifested by the relative locations in seismicity imaging, the relocation process can significantly improve the imaging resolution (e.g., Lomax & Savvaidis, 2022; Trugman & Shearer, 2017; Waldhauser & Ellsworth, 2000). Notably, the differential time (dt) data used in hypoDD can come from catalog picks (CT, i.e. dt.ct) or cross-correlation (CC, i.e. dt.cc), and they can be jointly inversed or used individually. The best practice of using dt.ct and dt.cc comes from their different characteristics: dt.ct covers a larger inter-event distance, whereas the precision is relatively low, because it is dependent on the phase picking and location

accuracy; dt.cc realizes a sub-sampling-rate measurement for the differential time and is unaffected by picking errors, but it covers a much smaller distance, because it relies on waveform similarity between event pairs (as summarized in Waldhauser, 2001). Thus, jointly using dt.ct and dt.cc in some way is usually suggested. In our AI-PAL workflow, we provide two approaches of combining dt.ct and dt.cc (Figure 1): (a) sequentially relocate with only dt.ct and dt.cc, and (b) relocate with dt.ct in the first step and jointly inverse dt.ct and dt.cc in the second relocation. The first approach is suitable for a relatively dense seismic network, because the near-source stations tend to have higher CC values; the second approach can maintain more events under a relatively sparse network, while also take advantage of the high-precision dt.cc data.

The calculation of differential times is the fundamental step for hypoDD relocation. The hypoDD software (Waldhauser, 2001) provides a ph2dt module that calculate the dt.ct data from the input phase file. It forms a chain of dt-links between events by searching neighboring events within a certain radius. For each event, a maximum number of neighboring events are preset in this process, in order to lower down the computational complexity and model errors. Following a similar strategy, we develop a ph2dt_cc module to calculate the high-precision dt.cc data with waveform cross-correlation (Figure S9 in Supporting Information S1). It first finds all possible event pairs by comparing the location differences and the common station picks. For each event pair, only events within a certain hypocentral separation and with certain number of shared stations are selected as candidates. To control the quality of dt measurements, we only use stations within an epicentral distance of about 100 km, and too small events (e.g., $M_L < 0$) cannot be linked to each other. To avoid too many measurements, we also limit the maximum number of stations for each event pair, and each event can only be linked to a maximum number of neighbors. Second, we calculate the CC-derived dt for each candidate event pair. In the CC calculation, users can set the window length, which channel to use, and the filtering frequency band. The weight for each phase used in the hypoDD is determined by the square root of CC value. After the CC calculation, we further select the dt measurements by discarding that with a too large dt or a too small CC, and the minimum number of station criteria still applies afterward. Detailed parameters will be given for each real case in the following sections.

2.2. Self-Attention RNN (SAR) Model for Phase Picking

The performance of a deep learning model is decided by multiple factors, including the model structure, target function (labeling strategy), training data, and training parameters. Here, we introduce a new phase picking neural network stemming from our previous work (Zhou et al., 2019), but incorporates the recent advances in this field targeting at the above factors (Figure 2).

2.2.1. Model Architecture

We adopt a Self-Attention RNN (SAR) model for phase picking. Recurrent Neural Network (RNN) is a typical deep learning model for sequence modeling, and has been widely adopted in various tasks related to time series, including speech recognition (e.g., Graves & Jaitly, 2014; Hannun et al., 2014) and natural language processing (e.g., Cho et al., 2014; Sutskever et al., 2014). Compared with Transformer (Vaswani et al., 2017), which is recently proven to be powerful in sequence-related tasks, RNNs have much fewer parameters, thus are more data-efficient and require less computational resources, making them a practical choice for small datasets (Ezen-Can, 2020; Peng et al., 2023). RNNs process sequences in a recurrent manner: generating outputs for each time step by integrating the current input with a hidden state that captures previously encountered sequence information. Zhou et al. (2019) first adopt a two-layer bi-directional Gated Recurrent Unit (GRU) RNN in the phase picking task, where the RNN is trained to classify each 1 s time step as noise, *P*-wave, or *S*-wave. In this study, we adopt the same GRU structure with two bi-directional layers, and 128 hidden size, but divide the raw data into frames of a much smaller step length (0.5 s) and stride (0.1 s) to increase the theoretical phase picking precision (Figure 2a). Moreover, we add a multi-head self-attention layer (Bahdanau et al., 2014; Vaswani et al., 2017) after the GRU layers (Figure 2a).

Multi-head self-attention is a mechanism that enhances the performance by allowing the model to simultaneously focus on different parts of the input sequence from different representation subspaces, which makes a key component of the Transformer architecture (Vaswani et al., 2017). In a single self-attention head, the input sequence (i.e., the output of GRU layers in our case) is transformed into three vectors: queries (*Q*), keys (*K*), and values (*V*), which are then used to compute attention scores that determine how much focus each element of the input sequence should have on every other element. This process captures dependencies regardless of their distance in the sequence, making the model able to capture complex dependencies that span across long

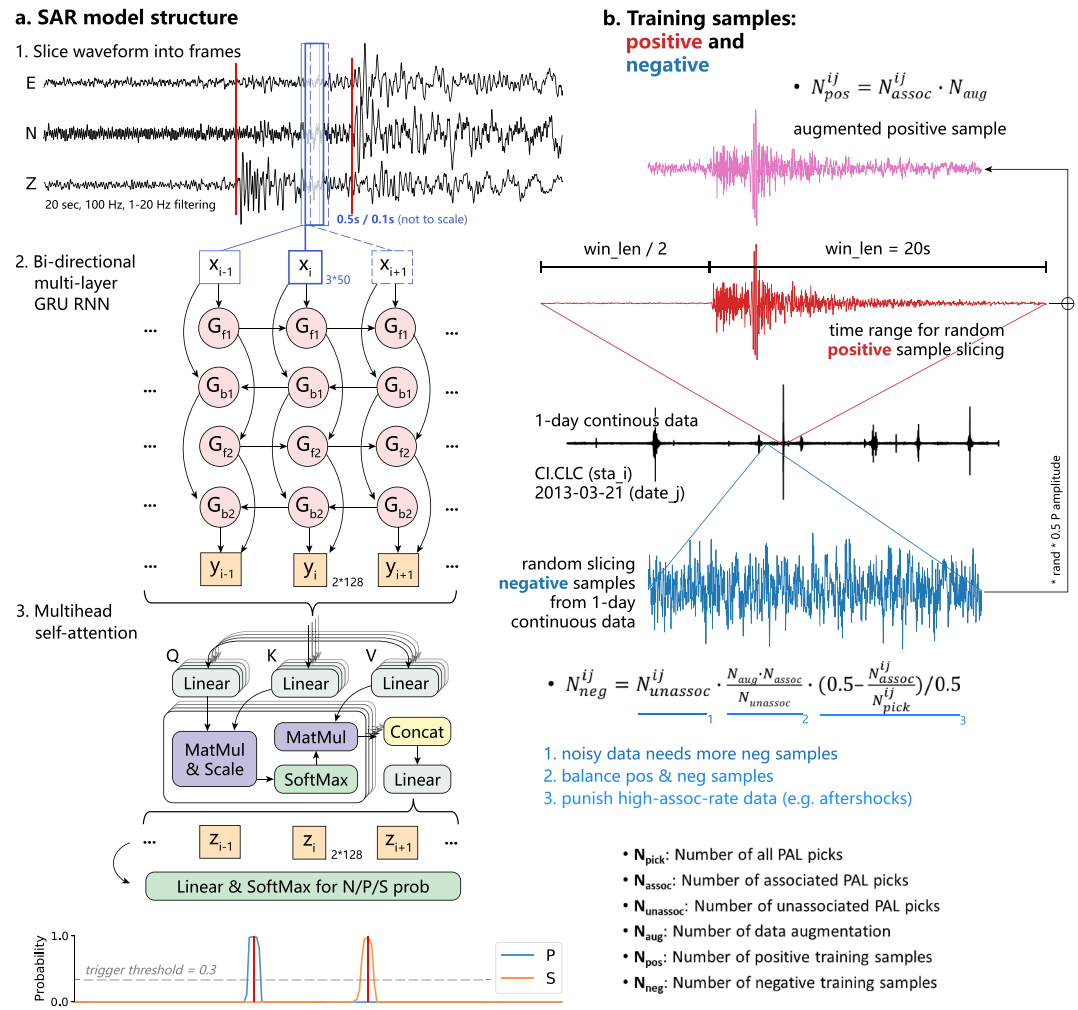


Figure 2. The Self-Attention RNN (SAR) model structure and training sample slicing strategy. (a) SAR model structure. The filtered 3-channel seismogram is plotted in black curves, and the red vertical lines mark the *P* & *S* arrival times. The data processing units are denoted by: x_i for the i th input time step; G_f and G_b for forward and backward Gated Recurrent Unit (GRU); y_i for the i th output of the GRU RNN; $Q/K/V$ for Query, Key, and Value input in the multi-head self-attention terminology, and they are all set as the Recurrent Neural Network output (i.e. y) in the case of self-attention; “Linear” for fully connected layer; “MatMul” for matrix multiplication; “Concat” for concatenation; z_i for the i th output of the multi-head self-attention layer; “N/P/S prob” for Noise, P wave, and S wave prediction probability, and (b) training sample slice strategy. Blue, red, and pink waveforms for negative, positive, and augmented positive samples, respectively.

sequences. Multi-head self-attention improves upon this by dividing the Q - K - V vectors into multiple independent heads, performing the self-attention process parallelly. Thus, each attention head learns to focus on different features of the input sequence, allowing the model to capture a richer array of relationships within the data. The outputs of all attention heads are then concatenated and linearly transformed to produce the final output, which combines the diverse learned representations. Similar attention mechanisms have been proved effective in enhancing the seismic phase picking performance (e.g., Mousavi et al., 2020; Zhang, Li, & Zhang, 2023). In this study, we employ four independent attention heads to balance the model complexity and the size of training sets. This leads to about 600,000 trainable parameters, which is of a similar complexity as PhaseNet (Zhu & Beroza, 2018).

For the model target in the training stage, we label only the frames containing *P* and *S* arrivals as *P* and *S*, respectively, and all other frames as Noise (Figure 2a). This labeling strategy forms a small but finite weight for the *P* & *S* arrival times, which guides the model to focus on these features and makes more stable phase detection. Since the invention of such labeling strategy by Zhu and Beroza (2018) for PhaseNet, it has been widely adopted

by most of the following deep learning models (e.g., Mousavi et al., 2020; Sun et al., 2023; Yu & Wang, 2022). In the model prediction stage, instead of treating the SAR output as classifications, we only use the output prediction probability, so that users can set the detection threshold based on their own problems. In line with PhaseNet, we also set the default triggering threshold for SAR as 0.3 to balance the detection completeness and accuracy. We regard a group of consecutive frames with a prediction probability above the threshold as a P or S pick, and take the median time of these frames as the picked phase arrival. For the cases when multiple P and S picks exist in a sliding window, we group these picks into all possible $P\&S$ pairs, only requiring that the S wave always arrives later than the P wave in a pair. Also, since we apply SAR in a sliding window manner, and that the sliding windows have about half the length overlapping, we will merge the picks from different windows if they have similar $P\&S$ picks.

2.2.2. Sampling Strategy for Training Set

The training samples fundamentally determine how the model behaves. To train the SAR model with the local earthquakes detected by PAL, we design a sampling strategy to properly balance the model's ability of detecting weak signals and identifying different types of noises (Figure 2b). We slice both positive (i.e., earthquakes) and negative (i.e., noise) samples for training. The positive samples are sliced surrounding the associated $P\&S$ arrivals (Figure 2b). We randomly position the P arrival at the first half of the time window, so that the model has a consistent detection ability for random signal positions that it will encounter in the real applications. Data augmentation was employed to increase the diversity of the training set and improve the generalizability of the SAR model. This involves adding real noise, randomly sliced from the same station-date for a specific $P\&S$ pick. For each augmented sample, the noise amplitude is scaled by a random ratio between 0 and 0.5, multiplied by the maximum P -wave amplitude. In this way, the number of positive samples is:

$$N_{\text{pos}}^{ij} = N_{\text{assoc}}^{ij} \cdot N_{\text{aug}}, \quad (1)$$

where i and j is the station and date index, respectively; N_{pos} is the number of positive samples; N_{assoc} is the number of associated picks; N_{aug} is the number of augmentations set by the users. Note that the users need to set the number of augmentations, so that the number of positive samples is large enough (empirically, >100,000). The negative samples are sliced randomly in each station-date pair (Figure 2b), while excluding the time ranges that has PAL-picked $P\&S$ arrivals. The number of negative samples on a certain station-date is decided by the number of associated and unassociated PAL picks on that station-date:

$$N_{\text{neg}}^{ij} = N_{\text{unassoc}}^{ij} \cdot \frac{N_{\text{aug}} \cdot N_{\text{assoc}}}{N_{\text{unassoc}}} \cdot \frac{0.5 - \frac{N_{\text{assoc}}^{ij}}{N_{\text{pick}}^{ij}}}{0.5}, \quad (2)$$

where i and j is the station and date index, respectively; N_{neg} is the number of negative samples; N_{pick} , N_{assoc} and N_{unassoc} are the number of all picks, number of associated and unassociated picks, respectively; N_{aug} is the number of augmentations set by the users. The logic for each term of this equation is that: (a) the number of unassociated PAL picks indicates the noise level, the more noise a station-date is, the more negative samples we need to slice on it; (b) the number of positive and negative samples need to be balanced in the training process; (c) the pick association ratio indicate the true positive rate of the PAL picks, and a large number of unassociated picks and a high association rate can occur together in an intense sequence (e.g., aftershock sequences), thus we should reduce the number of negative samples in this case. Note that if the association ratio is larger than 0.5, we do not slice any negative samples on that station-date. In this way, we can obtain a temporally stable number of negative samples (Figure S7 in Supporting Information S1). We will show in this study that the above sampling strategy realizes stable detection performance.

2.2.3. Model Training

The training of SAR is performed in a mini-batch manner. We feed both positive and negative samples in each iteration, and train for about 15–20 epochs to make about 100,000 total training iterations. Note that with the third term of Equation 2, the total number of negative samples will be smaller than the positives, thus we feed 128 positives and a smaller number of negatives so that they experience the same number of epochs. We adopt the

Adam optimizer (Kingma & Ba, 2014) with a learning rate of 10^{-4} . As will be shown in the real cases here and in our previous experiments for RNN (Zhou et al., 2019), the SAR model behaves very stable in the training process, without showing any signs of overfitting. In the following context, we may also refer to our workflow as SAR-PAL, since different neural networks adapt to different training sets, and we are only proving the effectiveness of the SAR-PAL combination in this study.

3. Test Datasets and Catalog Building

To test the detectability and generalizability of our AI-PAL workflow, we apply it to two cases that differ in tectonic settings, spatiotemporal scales, and network configurations: (a) the Ridgecrest-Coso (California) region from 2008 to 2019/07, covering its long-term preseismic period and 20 day's early aftershocks; and (b) the East Anatolian Fault Zone (EAFZ, Turkey) from 2020 to 2023/04, which covers ~ 3 years' preseismic period and ~ 3 months' aftershocks. Both cases contain large earthquakes that significantly change the seismicity rate and patterns, which makes the cataloging more difficult, and is thus suitable for technical discussions. In these cases, we compare AI-PAL with other popular cataloging workflows in terms of earthquake detection accuracy, completeness, and stability.

3.1. Case 1: Ridgecrest-Coso Region (2008–2019/07)

3.1.1. Background and Motivation

The 2019 Ridgecrest, California sequence comprised a M_w 6.4 foreshock on 2019/07/04, followed by a M_w 7.1 mainshock on 2019/07/06, and was featured by intense aftershock activities that reveal widespread orthogonal structures (Ross, Idini, et al., 2019; Shelly, 2020). It is associated with a young fault system (Goldberg et al., 2020; Hauksson & Jones, 2020; Xu et al., 2020) and the majority of the ruptured faults are not mapped before the earthquake (Thompson Jobe et al., 2020). Tectonically, the Ridgecrest sequence is situated within the East California Shear Zone (ECSZ), an approximately 100 km-wide band dominated by right-lateral strike-slip faults that accommodate the relative motion between the Pacific and North American plates (Oskin et al., 2008; Spinler et al., 2010; Tymofeyeva & Fialko, 2015). In this study, we focus on an area not only covering the Ridgecrest ruptures, but also two close-by tectonic units (Figure 3): (a) the central Garlock fault (Ganev et al., 2012; Hatem & Dolan, 2018; McGill & Sieh, 1993), a major fault cutting off the Ridgecrest faults on its south; and (b) the Coso Geothermal Field, one of the largest three geothermal fields in California that has been inducing intense seismicity for decades (Im et al., 2021; Schoenball et al., 2015; Trugman et al., 2016).

We work on a time period from 2008 to 2019, because of the significant improvement in the seismic network starting from 2008 (Hutton et al., 2010), and to overlap with the decade-long quake template matching (QTM) catalog that covers 2008–2017 (Ross, Trugman, et al., 2019). This time frame offers an exceptional opportunity to directly compare our novel workflow with MFT. For the aftershock period, we also align our investigations with the MFT catalog by Ross, Idini, et al. (2019), which spans the first 20 days since the M_w 6.4 foreshock. Both time periods of the QTM catalog utilize the routine SCSN catalog as templates, adopting a detection threshold of 12 times the median absolute deviation of CC, and are relocated with Growclust (Trugman & Shearer, 2017) that inverse the CC-based differential times. Also involved in the comparison is the relocated SCSN catalog (Hauksson et al., 2012), which has a similar number of detections as the routine SCSN catalog, but has a much higher relocation precision that comes from the waveform cross-correlation. To make a fair comparison, we utilize the same set of stations maintained by SCSN, which is composed of broad-band and short-period stations of three- and one-component recordings, along with several near-source temporary stations deployed after the 2019 mainshock (Figure 3 & S1a-b).

3.1.2. Detection and Location

For this case, we run the AI-PAL workflow for the ~ 11 year preseismic period and the 20 day aftershock period separately, due to the considerable differences in station distribution (Figure 3 and Figure S1 in Supporting Information S1) and seismicity rate. This approach also allows for an assessment of the workflow's performance under two end-member situations: a scenario of very intense sequence observed by a rather dense seismic network, and a pure interseismic period (no large earthquakes) with regular observational condition. We set the SAR-PAL parameters specifically for the preseismic period, since it is less investigated; and we keep the same set of parameters for the aftershock period to make the results more comparable.

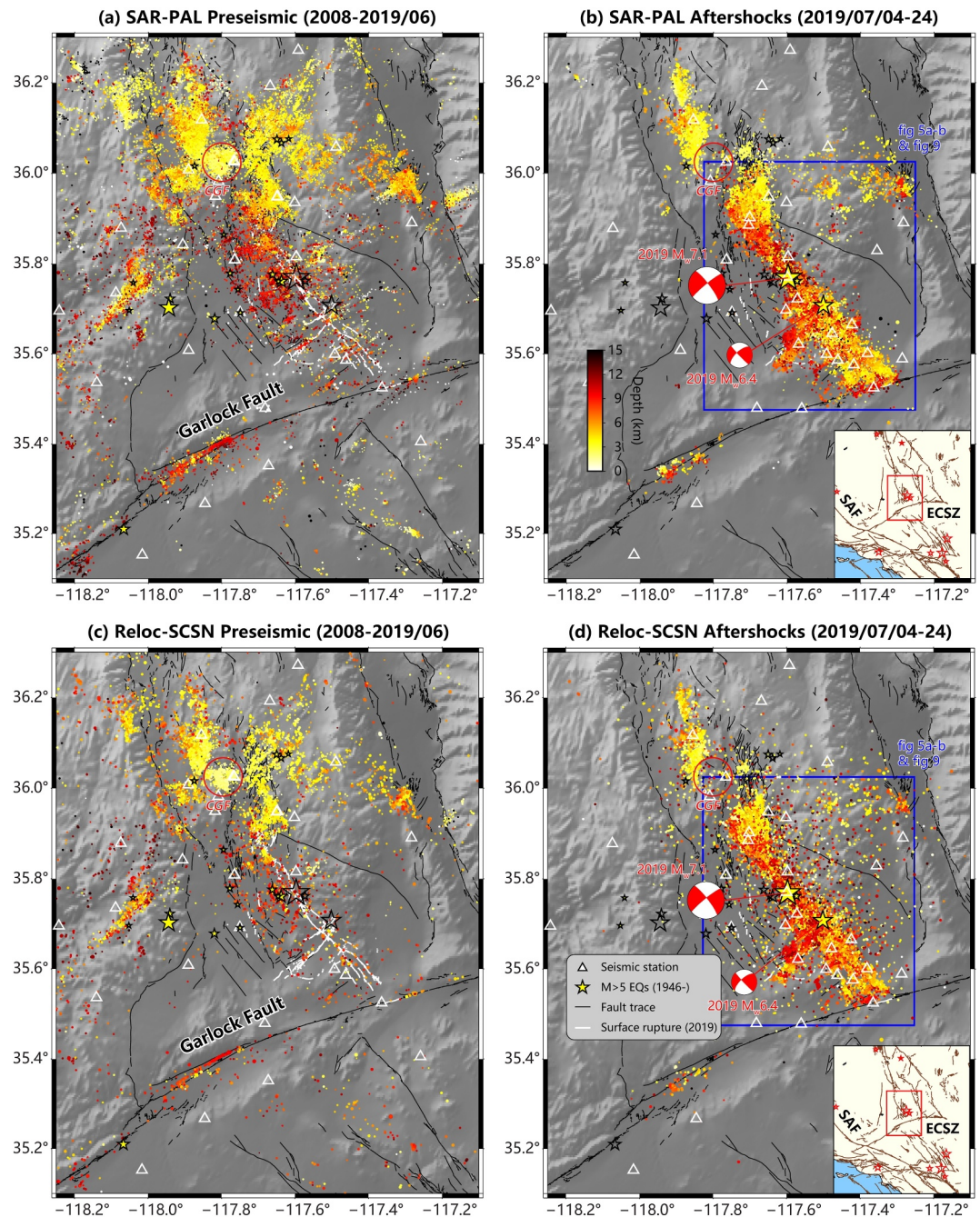


Figure 3. Study region and catalog comparison for Case 1: Ridgecrest-Coso (2008–2019/07). (a)–(b) Plot the SAR-PAL catalog generated in this study, with preseismic period (2008–2019/06) and aftershock period (2019/07/04–24) shown separately. Seismicity is plotted as dots that have its depth denoted by color and size varies by magnitude. $M > 5$ earthquakes after 1946 are marked by yellow stars. Seismic stations used are denoted by white hollow triangles; active faults are plotted as black lines; surface rupture caused by the 2019 Ridgecrest earthquakes are marked by white lines. The area of Coso Geothermal Field is marked by a red circle. The insets show location of the study area in a larger scale, with the San Andreas Fault and Eastern California Shear Zone (ECSZ) marked, and (c)–(d) are the same as (a)–(b), but for the relocated Southern California Seismic Network catalog (Hauksson et al., 2012). The blue box in (b) and (d) marks the location coverage in Figures 5a and 5b and 9.

In running PAL, we set the STA/LTA triggering threshold as 12 (defined by energy), S wave searching window as 0–12 s after the P pick, in line with the scale of study region and the average inter-station distance. For PAL association, we require at least four stations to have an original time deviation < 1.2 s, and a maximum P-wave

travel time residual <0.8 s. The PAL obtains 61,053/451,694 and 49,737/440,942 event detections/associated picks for the preseismic and aftershock period, respectively.

In running SAR, we set the window length as 20 s and the sliding stride as 10 s in the predictions for continuous data. We augmented the original training samples by two times, making a total of 812,771/451,689 positive/negative samples for training in the preseismic period, and 792,784/123,018 positive/negative training samples for the aftershock period. Note that there are $\sim 10\%$ remaining samples serve as the validation set. The waveforms in a time window are band-pass filtered to 1–20 Hz and normalized in both the training and application stage. We made 15 epochs of training until the accuracy is stable. As described in the last section, we apply the local-trained SAR picker and the same PAL associator to enhance the PAL detections. This gives 165,393/1,277,974 and 122,933/1,078,038 event detections/associated picks for the preseismic and aftershock period, respectively. In summary, for the Ridgecrest-Coso region, the AI-PAL workflow achieves a ~ 2.5 – 2.7 -fold increase in detection number compared to PAL.

To locate and relocate the SAR detections, considering a rather dense network, we adopt the first approach described in Section 2.1 (Figure 1), utilizing dt.ct and dt.cc separately for a two-step relocation. Similar as the detection part, we adopt the same set of location parameters for both the preseismic and aftershock period.

For the hypoINV absolute location, the distance weighting is configured as follows: 0–50 km assign full weight, >100 km zero weight, and 50–100 km is assigned a cosine tapered in between. A similar weighting scheme is employed for the time residual between 0.25 and 0.75 s. For S waves, the weights are further adjusted by a factor of 0.6, considering a larger picking error relative to P waves. The above weighting strategy results in 95.4% events well located, and an average lateral and vertical uncertainty of 1.11 and 2.22 km, respectively.

In the hypoDD relocation process with dt.ct, we pair events within 8 km (i.e., WDCT, in hypoDD terminology), P and S waves weighted respectively as 1 and 0.6, and inverse for four iterations. This results in a relative location error of about 120 m for epicenter and 150 m for depth, under the least squares criteria by hypoDD. Given that the least-square location error reported by hypoDD tends to underestimate the real uncertainty (Waldhauser, 2001), we test the dt.ct relocation results under different velocity models, and found highly consistent distributions (Figure S15 in Supporting Information S1).

In the final CC relocation stage, we measure the waveform-based differential time on 1–20 Hz band-pass filtered waveforms, with the P and S window lengths set as 2.5 and 4 s, starting from 0.5 to 0.2 s before the phase arrival, respectively. We do not calculate dt between $M_L < 0$ earthquakes to lower down the computational costs. A maximum number of 200 neighboring events are preset in the candidate neighbor selection step, sorted by the separation distance. After the CC calculation, we discard the measurements with $CC < 0.35$, or $dt_p > 0.5$ s, or $dt_s > 0.8$ s, and the event pairs with <4 stations fulfilling these criteria are further dropped. The above selections finally maintain 18,421,488 P and 12,506,600 S differential time measurements for the preseismic period. For the aftershocks, these numbers are 13,718,530 for P and 8,422,550 for S .

In the hypoDD relocation process, we link events within 4 km (i.e., WDCC, in hypoDD terminology), set both P and S waves' weights as 1 and inverse for four iterations, because S waves have higher stability in waveform cross-correlation. Finally, we obtained 101,193 and 61,578 well-relocated events for the preseismic and aftershock period, respectively. For both catalogs, the average relocation error reported by hypoDD is about 40 and 60 m along horizontal and vertical directions, respectively. Note that this location error is in the relative and average sense, and reflects more about the goodness of data fitting, instead of the real uncertainty.

The SAR-PAL catalog and the relocated SCSN catalog exhibit an overall consistent distribution, particularly in areas of major intense clusters. Nonetheless, the SAR-PAL catalog demonstrates an enhanced detection capability, rendering previously indistinct seismicity structures much more discernible (Figure 3).

3.2. Case 2: East Anatolian Fault Zone (2020–2023/04)

3.2.1. Background and Motivation

On 6 February 2023, Southeast Turkey's Kahramanmaraş region was struck by an M_w 7.8 earthquake followed by another M_w 7.6 within ~ 9 hr, marking one of the largest continental earthquake doublets ever documented (Hussain et al., 2023; Jia, 2023; Ren et al., 2024; Xu et al., 2023; Zhang, Tang, et al., 2023). The 2023 Turkey earthquake doublet occurs on the East Anatolian Fault Zone (EAFZ), a roughly ~ 600 km-long fault zone defining

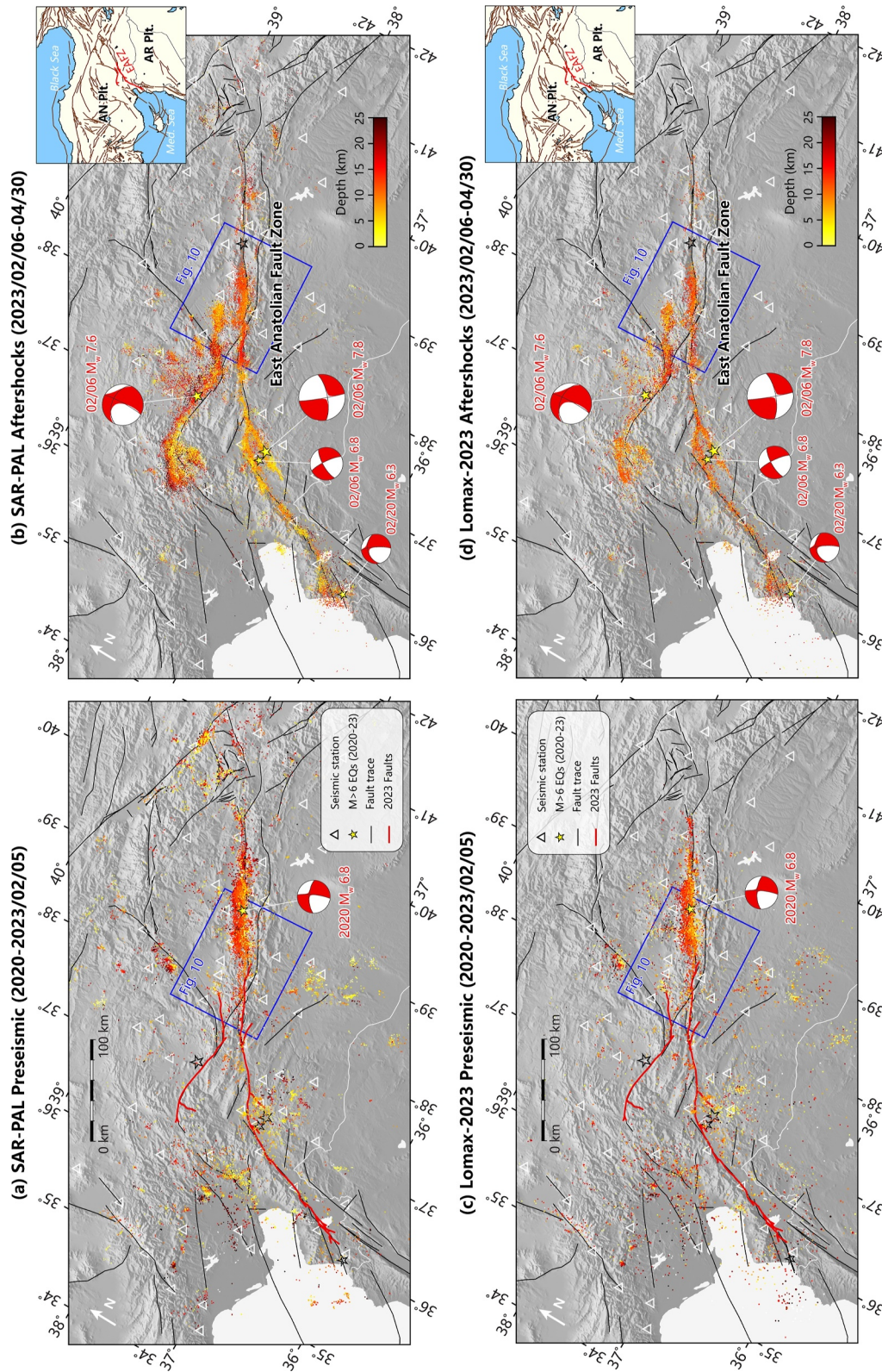


Figure 4. Study region and catalog comparison for Case 2: East Anatolian Fault Zone (2020–2023/04). (a–b) Plot the SAR-PAL catalog generated in this study, with preseismic period (2020–2023/02/05) and aftershock period (2023/02/06–04/30) shown separately. Seismicity is plotted as dots that have its depth denoted by color and size varies by magnitude. $M > 6$ earthquakes after 2020 are marked by yellow stars. Seismic stations used are denoted by white hollow triangles; active faults and responsible faults for the 2023 doublet are plotted as black and red lines, respectively. The insets show tectonic background, with the Arabian Plate (AR Plt.) and Anatolian Plate (AN Plt.) marked, and (c–d) are the same as (a–b), but for the relocated AFAD catalog by Lomax (2023). The blue boxes mark the location coverage in Figure 10.

the plate boundary between the Anatolian and Arabian plates (Figure 4). Driven by the collision between the Arabian and Eurasian plates, the Anatolian plate is extruded westward at a rate of approximately 20–25 mm/yr, along with a counterclockwise rotation (Barbot & Weiss, 2021; Bulut et al., 2012; Güvercin et al., 2022; McClusky et al., 2000). This plate motion gives rise to the predominantly left-lateral strike-slip characteristics observed along the EAFZ. Additionally, the fault slip rate along the EAFZ is relatively low, ranging from ~4 mm/yr on the SW segments to ~10 mm/yr on the central and NE segments (Aktug et al., 2016; Cavalié & Jónsson, 2014; Walters et al., 2014; Weiss et al., 2020). It is noteworthy that the two mainshocks of the 2023 doublet are associated with different fault systems (Figure 4): the first M_w 7.8 mainshock occurs on the major plate boundary faults, whereas the second M_w 7.6 mainshock ruptures the intraplate faults. The distinct aftershock patterns observed across this dual fault system suggest contrasting fault properties (Ding et al., 2023; Güvercin, 2024).

In this study, we build a seismic catalog covering the entire EAFZ, a task that presents technical challenges due to the large spatial extent, and the relatively sparse seismic network, with its inter-station distances ranging from about 30 to 60 km (Figure 4 and Figure S1c in Supporting Information S1). This network density is representative of most regions outside of the well-monitored areas like California and Japan. We start our analysis from 2020, because of the significant enhancement of Turkey seismic network (Figures S1c and S2b in Supporting Information S1) spurred by a M_w 6.8 earthquake on 24 January 2020, which ruptured the NE side of the 2023 rupture zones (Galović et al., 2020; Lin et al., 2020; Melgar et al., 2020; Pousse-Beltran et al., 2020). Our study extends over a 3 month aftershock period to 2023/04, which expands our previous rapid work that only covers 1 month' early aftershocks (Ding et al., 2023) and the 2 month aftershock period examined by Güvercin (2024). We combine 3-channel broad-band stations from multiple networks, including the TU network from the Disaster and Emergency Management Presidency (AFAD) and the KO network from the Kandilli Observatory and Earthquake Research Institute (KOERI). We will combine both the case in the Ridgecrest-Coso and this EAFZ case to evaluate the generalization ability of different phase picking and association algorithms in earthquake detection. This is motivated by the fact that the two cases are associated with contrasting tectonic settings, spatiotemporal scales, and network configurations.

3.2.2. Detection and Location

Contrary to our first case in Ridgecrest-Coso, for this case in EAFZ, we execute the AI-PAL workflow over a combined ~3.5 year pre-postseismic period. This approach offers additional validation of AI-PAL's capability to handle highly variable seismicity rates. The detection and location parameters for this EAFZ case largely align with those used in the Ridgecrest-Coso case, with the exception of several parameters adjusted to accommodate the expanded scope and density of the network.

In running PAL, the phase picker utilizes the same triggering threshold, but an 18 s S wave search window, which corresponds to a ~150 km epicentral distance. The PAL associator maintains a 4-station requirement, but differ in the limit for original time deviation as <1.6 s and the P-wave travel time as <1.2 s. These specifications resulted in 47,416/295,494 event detections/associated picks. In running SAR, we set a larger window length of 25 s and maintain the same 10 s sliding stride. We augment the original training samples by 2 times, making a total of 516,168/322,894 positive/negative training samples. Similarly, ~10% remaining samples are used for validation. We applied the same data preprocessing, and made 20 epochs of training until the accuracy is stabilized. After applying the locally trained SAR picker and the same PAL associator, we got 140,119/929,944 event detections/associated picks, which represents a ~3-fold increase in the number of detections compared to PAL.

To locate and relocate these detections, we adopt the second approach described in Section 2.1 (Figure 1), utilizing both dt.ct and dt.cc in the second round of relocation. For the hypoINV absolute location, we also weight the phases by their epicentral distance and residual times, but set a less restrictive set of parameters to account for the sparser network: the distance weighting is configured as 0–80 km full weight, >160 km zero weight, and 80–160 km cosine tapered; the time residual weighting is configured in the same way between 0.4 and 1.2 s. Similar as for Ridgecrest-Coso, the S-wave weights are further adjusted by a factor of 0.6. The hypoINV outputs ~93.1% well-located events, with an average lateral and vertical uncertainty of 2.48 and 3.73 km, respectively. Note that the location error is highly variable across our study region, due to the heterogenous station distribution.

In the first round of hypoDD relocation with dt.ct, we pair events within 20 km (i.e., WDCT, in hypoDD terminology), and P and S waves weighted as 1 and 0.6, inversed for four iterations. The relocation uncertainty is

also correlated with the station density, ranging from about 180 to 280 m for epicenter and 250–550 m for depth, under the least squares criteria by hypoDD.

For CC-based differential time measurements, we filter the waveforms to 1–12 Hz, considering the sparser network, while keeping the same *P* and *S* windows as in Ridgecrest-Coso. We only calculate *dt* between $M_L > 0.5$ earthquakes and other events. We limit a maximum number of 50 neighboring events in the candidate selection step, and we require $CC > 0.3$, $dt_p < 1$ s, and $dt_s < 1.75$ s, in addition to a minimum number of 4 stations fulfilling these criteria. The above processes finally obtain 3,719,995 *P* and 2,961,716 *S* differential times.

For the final hypoDD relocation process that combines *dt.ct* and *dt.cc*, we made 2 sets of inversions. In the first round, we set WDCC and WDCT as 4 and 10 km, and weighted the *dt.ct* data strongly; in the second round of inversion, the WDCC and WDCT are set to 2 and 5 km, with the *dt.cc* being dominantly weighted. Using this weighting scheme, we benefit from both the high-precision *dt.cc* data and the wide coverage of *dt.ct* that helps maintain more events in relocation. Finally, we obtained 93,680 well-relocated events, with an average relocation error ranging from about 40 to 80 m and 70–120 m along horizontal and vertical directions, respectively.

Furthermore, aiming for a higher-resolution catalog (at the cost of reducing the number of detections), and for a more realistic estimation of location uncertainty, we also applied the first approach outlined in Section 2.1 (Figure 1) to relocate the SAR detections. We exclusively used *dt.cc* for relocation, setting WDCC as 2 km and performing the inverse over four iterations. This yielded 56,656 events with a much lower relocation uncertainty and an overall consistent distribution of seismicity (Figures S19–S21 in Supporting Information S1). However, it is important to note that this relocation approach results in a considerable reduction of events above the completeness magnitude, rendering it less suitable for *b*-value analysis.

The detection and location strategies elaborated above result in an enhanced catalog compared with the relocated AFAD catalog produced by Lomax (2023), as shown in Figure 4. Such advancement is made possible by a much higher detectability and relocation precision.

4. Comparisons of Cataloging Workflows

4.1. Comparison With Matched Filter

We compare AI-PAL with the MFT using the Ridgecrest-Coso case, as the recognized matched filter catalog QTM covers the same spatiotemporal range (Figure 3). Specifically, we compare the frequency-magnitude distribution (FMD) and magnitude-time sequences of the relocated SCSN, the QTM, and the SAR-PAL catalog for both the aftershock and preseismic period (Figure 5).

4.1.1. Comparing Frequency-Magnitude Distributions

FMDs serve to illustrate not only the total number of detections, but also the distribution of events across different magnitude bins. Ideally, a FMD should adhere to the GR law (Gutenberg & Richter, 1944), meaning that for the magnitude range above a completeness threshold (approximately the magnitude of maximum non-cumulative distribution, Wiemer & Wyss, 2000), the occurrence frequency of earthquakes is expected to follow a power-law distribution relative to their magnitudes (i.e., a linear relationship when plotted on a logarithmic scale). In comparing FMDs, consistency in distributions above the magnitude of completeness indicates reliable detections, while deviations suggest a large portion of false or missed detections. The objective of earthquake detection is also to achieve the smallest possible magnitude of completeness, or equivalently, the largest possible cumulative count under a consistent slope. The reliability of the additional detections is evaluated by the ratio of well-located events.

Applying the above criteria to analyze the FMDs, we find that SAR-PAL provides a comparable or even superior detectability compared with QTM (Figures 5a–5c). Although QTM detected 102,820 aftershocks, compared to 78,934 by SAR-PAL (Figure 5a), it is important to note that ratio of well-located events determined by the Growclust algorithm falls below 35% (Figure 5b). This result also suggests a trade-off between the quantity of detections and the quality of their locations, emphasizing the necessity of keeping comparable relocation precision for an equitable comparison of detectability across methodologies.

In addition to overall event counts, the slope of these FMDs also offer critical insights. First, an apparent inconsistency in detection completeness exists in both the SCSN and QTM catalogs between the M 0–2 and $M > 2$

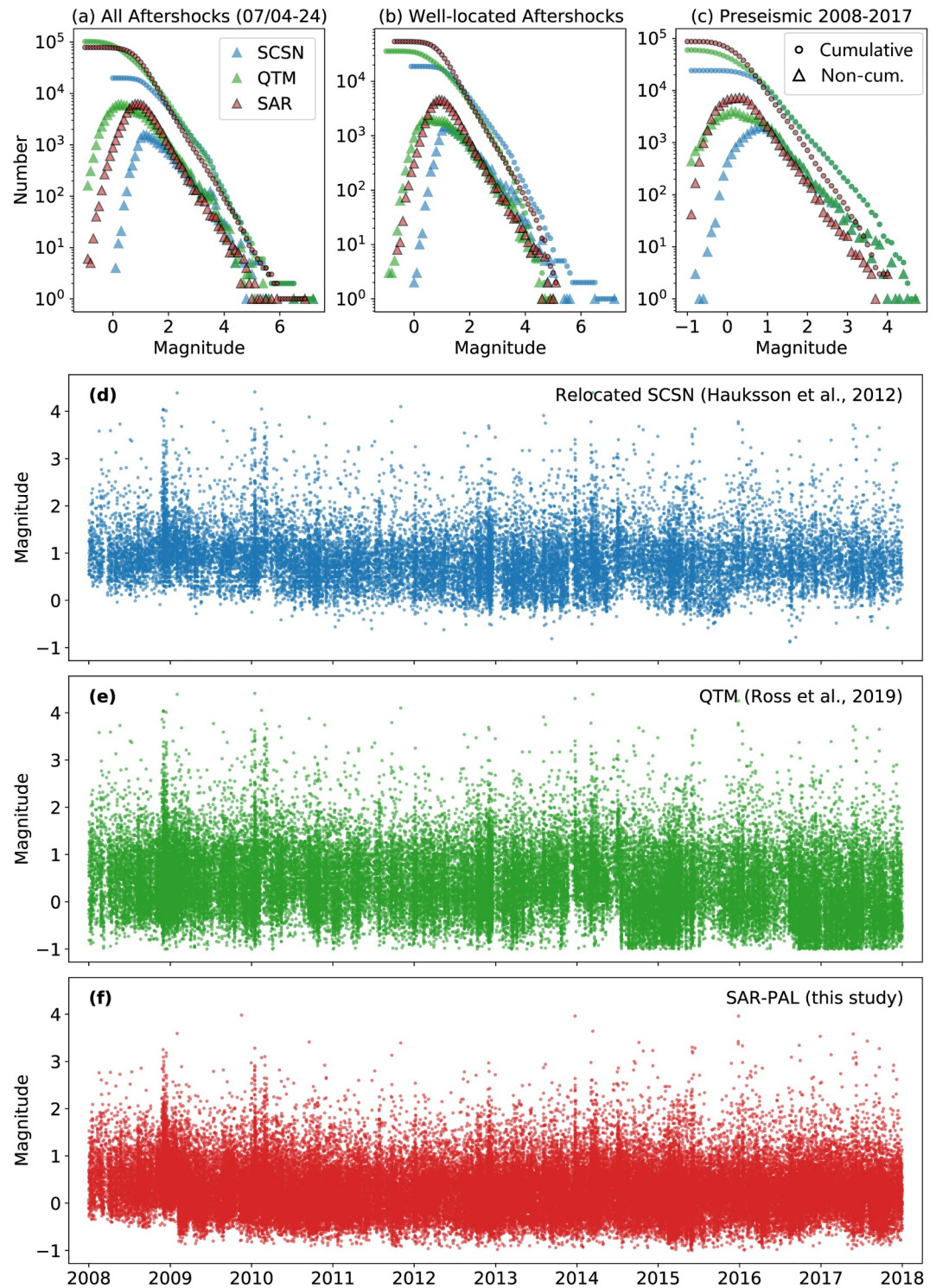


Figure 5. Comparison of frequency-magnitude distribution (FMD) and magnitude-time sequence. The blue, green, and red color denote the Southern California Seismic Network catalog (Hauksson et al., 2012; Hutton et al., 2010), the quake template matching catalog (Ross et al., 2019a, 2019b), Ross, Trugman, et al., 2019 and the SAR-PAL catalog in this study, respectively. (a–c) Plot the FMD comparison for the total aftershock detections, well-located aftershocks, and the preseismic period (2008–2017), respectively. The dots and triangles denote cumulative and non-cumulative distribution in FMD, and (d–f) plot the magnitude-time comparisons for the preseismic period (2008–2017). Note that only the events within the coverage of Figure 3 are considered.

events, as evidenced by the varying slopes in the FMDs (Figures 4a–4c). Second, also for the SCSN and QTM catalog, there is a noticeable shift in the FMDs' slope around M 3.5 (Figures 4a–4c). This shift is partially caused by the adoption of different magnitude scales (e.g., M_l , M_w , or M_{lr} , as detailed at <https://scedc.caltech.edu/eq-catalogs/change-history.html>), which can affect the catalog-based b-value studies. The SAR-PAL catalog does not exhibit the aforementioned issues, demonstrating a consistent detection performance across the magnitude range in this study.

4.1.2. Comparing Magnitude-Time Sequences

The magnitude-time sequence reveals the temporal evolution of seismicity, which is modulated by external sources, such as tectonic loading, as well as internal triggering between earthquakes (Hsu et al., 2024; Ogata, 1988; Zaliapin & Ben-Zion, 2020; Zhuang et al., 2002). Thus, statistical studies will require a catalog to have temporally stable detection capability for both the intense sequences and the background seismicity. In our case in the Ridgecrest-Coso preseismic period, we observe that the SCSN catalog is not temporally consistent, displaying high detectability mainly during the intense seismic sequences (Figure 5d). This pattern arises from the SCSN catalog's compilation process, which involves a semi-automated detection supplemented by manual inspections, particularly for intense seismic sequences (as detailed at <https://scedc.caltech.edu/eq-catalogs/change-history.html>). The issue of temporal inconsistency is not alleviated, but being exacerbated in the QTM catalog, as indicated by the highly variable lower magnitude limit (Figure 5e). It is probably due to the fact that earthquakes within an intense sequence tend to have high waveform similarity, making them easy to be detected by matched filter; Conversely, the background seismicity comes from a wider variety of faults and asperities, which inherently diminishing the effectiveness of matched filter. In contrast, our SAR-PAL catalog realizes markedly improved stability in detection throughout this 11 year period (Figure 5f), without showing any notable gaps in the detection of lower-magnitude events. This suggests that the SAR model is adept at capturing the statistical features of seismic events over a large spatiotemporal range.

4.1.3. Comparing Computational Efficiency

In our final comparison, we assess the computational efficiency of the SAR-PAL workflow against QTM. The entire process (including running PAL) is executed within ~ 7 days with one Nvidia GeForce RTX 2080 GPU card and 1 Intel Xeon E5-2695 CPU. Contrarily, the construction of the QTM catalog requires 200 Nvidia P100 GPU cards, with a runtime exceeding 60 days. Although the study area of QTM is ~ 20 times larger, there is still a 100-fold difference in speed. This significant difference is attributed to the computationally intensive nature of cross-correlation and that the linear increase in runtime with the addition of templates in matched filtering.

4.1.4. Summary

In conclusion, the AI-PAL workflow not only exhibits comparable detection capabilities and superior stability compared to the matched filter approach, but also significantly outperforms in computational efficiency. This makes it particularly advantageous for applications across extensive spatiotemporal scales.

4.2. Comparison With Pre-Trained AI Models

Utilizing both cases in the Ridgecrest-Coso preseismic period and the EAFZ pre-to-postseismic period, we conduct detailed comparisons against pre-trained PhaseNet (Zhu & Beroza, 2018) and GaMMA (Zhu, McBrearty, et al., 2022), both of which are considered as the state-of-the-art phase picking and association algorithms, and are being widely adopted in cataloging workflows (e.g., Zhang et al., 2022; Zhu, Hou, et al., 2022). Through these comparisons, we can evaluate the scalability of these algorithms across different tectonic environments, spatiotemporal ranges, and network densities.

4.2.1. Data Processing With PhaseNet and GaMMA

In executing PhaseNet and GaMMA, we adhere primarily to their default parameters and the behaviors anticipated for typical users, which are intended to be broadly applicable.

For PhaseNet, we use 30 s time windows that slide by 15 s steps. They are larger than what we set for SAR picker in the two cases, and should have enough coverage. The waveforms are high-pass filtered above 1 Hz. We keep

the default triggering threshold at 0.3. Picks from overlapping sliding windows are merged using a 0.5 s window, where the picks with higher prediction probabilities are selected. Note that PhaseNet pick *P* and *S* arrivals independently, and we pair them up to combine with the PAL associator. The phase pairing strategy is consistent with that used for the SAR picker; however, it necessitates manually setting an *S*-wave search window, which we keep the same as for PAL picker in the two cases.

For GaMMA, we set the criterion at a minimum of 8 phases to declare an event detection, aligning this requirement with the four-station criterion of the PAL associator that we applied in both the EAFZ and Ridgecrest-Coso cases. It is important to highlight that the association results from GaMMA do not necessarily ensure paired *P* and *S* phases for each station. However, we observed that directly utilizing the GaMMA outputs for hypoINV leads to a significant number of events unconstrained (Figures S10–S11 in Supporting Information S1). Consequently, we opted to refine the GaMMA outputs by excluding the stations with only single *P* or *S* picks and imposed a requirement for a minimum of four stations. This refinement achieved location qualities comparable to those obtained using the PAL associator.

4.2.2. Comparing Number of Associated Picks and Association Ratio

We first compare the number of associated picks and the association ratio (Figures 6a–b & e–f). We assume that the association process filters out most of the false picks, thus the number of associated picks represents the phase detection completeness, and the association ratio indicates the detection accuracy.

In the Ridgecrest-Coso case, all three combinations of AI-driven picker + associator setups (i.e., PhaseNet + GaMMA, PhaseNet + PAL, and SAR-PAL) achieved a similar number of associated picks, with an association ratio around 10% (Figures 6a and 6b). This performance markedly surpasses that of the STA/LTA algorithm, as represented by PAL, demonstrating the advanced picking accuracy and completeness afforded by neural networks that learn from waveform features. Note that PhaseNet is originally trained on Northern California data, thus is intrinsically optimized for the Ridgecrest-Coso case. Moreover, all three methods recalled ~97% events in the SCSN catalog, showing great consistency in event detection.

In contrast, in the EAFZ case, the number of associated picks by SAR-PAL surpasses that of PhaseNet + PAL by ~1.5 times, and ~2.5 times more than PhaseNet + GaMMA (Figures 6e and 6f). Furthermore, both PhaseNet + PAL and PhaseNet + GaMMA demonstrate significantly lower association ratios (5.9% and 3.3%, respectively) in comparison to SAR-PAL's 23.1%, as well as to their performances in the Ridgecrest-Coso case (~10%). This result indicates that PhaseNet is less tailored for the EAFZ data, and that GaMMA suffers from a much larger-scale network with sparse station distributions.

By comparing the FMDs (Figures 6c and 6g), it is evident that the three AI-driven picker-associator combinations show high consistency in Ridgecrest-Coso, as previously noted. However, PhaseNet + GaMMA displays noticeable discrepancies for the EAFZ case, missing a significant portion of events above the complete magnitude. Excluding this particular instance, the remaining seven catalogs across both cases display consistent FMDs for the portion above completeness magnitude, aligning with the expectations set by the GR law.

4.2.3. Comparing Number of Unassociated Picks

Lastly, we examine the number of unassociated picks across all stations (Figures 6d and 6h). Given the pre-supposition that a large portion of the unassociated picks represent false detections, the spread of unassociated picks across stations serves as an indicator of a phase picker's detection stability. In our analysis, we compare pre-trained PhaseNet and SAR-PAL, employing PAL (based on STA/LTA) as the baseline result. The rationale behind this comparison stems from the high sensitivity of STA/LTA algorithms to pulse-like noises (e.g., data glitches), which often results in a higher volume of unassociated picks, as shown by the low association ratio mentioned above. Consequently, when PhaseNet or SAR generate significantly more unassociated picks than PAL, it suggests a high false detection rate for low-SNR phases.

In both Ridgecrest-Coso and EAFZ, there is no correlation in the number of unassociated picks between PAL and the two AI-based pickers (Figures 6d and 6h), demonstrating the AI pickers' effective ability to discriminate against pulse-like noises. This performance stems from their design to analyze whole-waveform characteristics. However, even in the case of Ridgecrest-Coso (Figure 6d), a few stations were identified where PhaseNet produced a significantly higher number of unassociated picks compared to PAL. Such instability of pre-trained

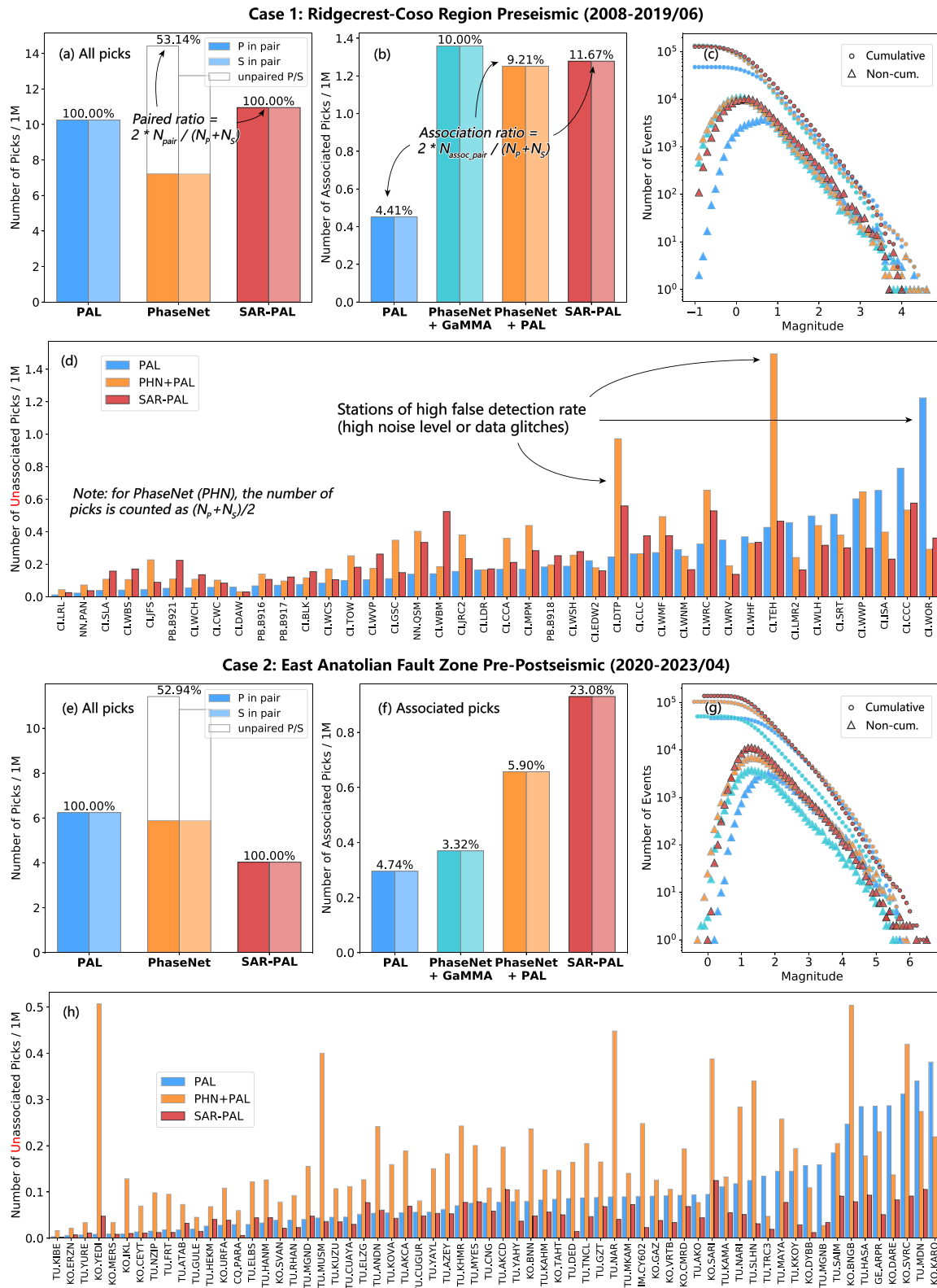


Figure 6.

PhaseNet is notably severer in EAFZ, where multiple stations recorded more than twice the unassociated picks than PAL (Figure 6h). These observations suggest that pre-trained PhaseNet may be sensitive to certain types of low-SNR noises. In contrast, such instable behavior is not obvious in SAR-PAL's picking results (Figure 6d). The above observations highlight a potential area for improvement in AI-picker's noise discrimination capabilities, and a multi-station-based picking algorithm could potentially offer a solution in the future (e.g., Feng et al., 2022; Sun et al., 2023).

4.2.4. Summary

In summary, our comparative analysis across cases with divergent characteristics demonstrate the SAR-PAL's superiority in detection completeness and stability over pre-trained PhaseNet, as well as PAL's greater scalability in phase association compared to GaMMA. These attributes position AI-PAL as a notably robust workflow for compiling long-term, large-scale seismic catalogs.

4.3. Comparison With Locally Trained AI Models

To further investigate whether the superior performance of SAR-PAL over pre-trained PhaseNet arises from its model structure or/and the use of a local training set, we conducted three additional experiments: (a) training the PhaseNet model using the PAL detections as described in Section 2.2 (referred to as the PHN-PAL catalog); (b) training the SAR picker with routine catalogs from SCSN and AFAD in the two cases (referred to as the SAR-SCSN/AFAD catalog); and (c) training the PhaseNet model using routine catalogs from SCSN and AFAD in the two cases (referred to as the PHN-SCSN/AFAD catalog). The evaluation strategy follows that outlined in the previous section.

4.3.1. Model Training

The training of all models in this section follows the same mini-batch strategy as described in Section 2.2.3. In building and training PhaseNet, we kept all hyper-parameters the same as originally designed in Zhu and Ber- oza (2018), except for the window length, which was adjusted to match that of SAR-PAL for fair comparison. In training with routine catalogs, we used SCSN phases for the Ridgecrest-Coso case, comprising 174,405 *P* and *S* pairs for 22,457 earthquakes occurring within the same spatiotemporal range; for the EAFZ case, we utilized AFAD phases, which include 263,437 picks from 35,773 events. In both catalogs, earthquakes were recorded by at least 4 stations, consistent with the detection criteria of PAL in this study. Since the routine catalogs do not provide negative samples, we retained the same negative set as sampled based on PAL detection results (see Section 2.2).

We visualize the training process by training curves (Figure 7), where the detection accuracy, defined as the ratio of correctly classified samples, is calculated on training and validation samples after each epoch of optimization. A positive prediction is made if both *P* and *S* phase picks are identified within the time window. Since the data and labels vary across training sets, we focus only on significant differences when comparing training curves. Furthermore, detection completeness is expected to be higher in real applications, as we adopt a threshold of 0.3 for continuous data instead of 0.5 during training. Finally, validation accuracy tends to exceed training accuracy because validation samples are not augmented with noise, unlike the training data.

In all cases, training accuracy stabilized after approximately 10 epochs' optimization, with classification accuracy for noise samples reaching nearly 100% (Figure 7). However, final earthquake detection accuracy varies across different model-data combinations and test cases. The PhaseNet model consistently exhibits 10%–30% lower training/validation accuracy compared to SAR when trained on the same dataset (Figure 7). Notably, in the EAFZ case, PhaseNet is not trainable with the AFAD dataset (Figure 7b), suggesting that the AFAD catalog likely

Figure 6. Comparison of phase picking and association performance for pre-trained models. The blue, cyan, orange, and red markers denote the results by PAL, PhaseNet picker + GaMMA associator, PhaseNet picker + PAL associator, and SAR-PAL workflow developed in this study, respectively. (a and e) plot the number of picks. The *P* and *S* picks are marked by darker and lighter colors. The paired *P*&*S* picks are filled by color, with the paired ratio annotated on top of the bar, (b and f) plot the number of associated picks. The pick association ratio is annotated on top of the bar, (c and g) plot the frequency-magnitude distribution comparison, where circles and triangles denote cumulative and non-cumulative numbers, respectively, and (d and h) plot the number of unassociated picks on each station. The stations are sorted by the number by PAL (a–d) and (e–h) plot the Case 1 (Ridgecrest-Coso) and Case 2 (East Anatolian Fault Zone), respectively. Note that for (c) and (g), only the events within the location coverage of Figures 3 and 4 are included in the statistics.

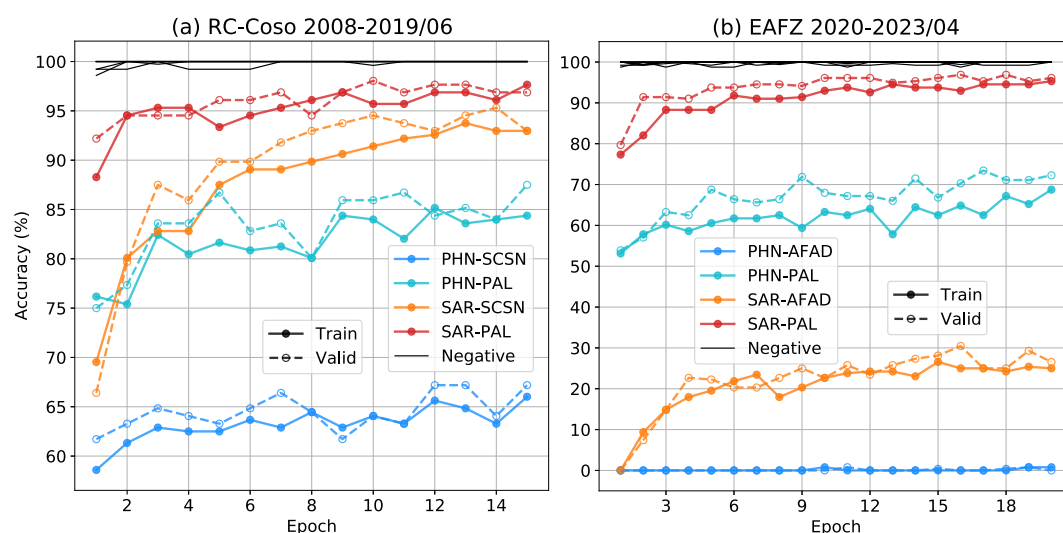


Figure 7. Training curves of different model-data combinations (a–b) Plot the training curves for the Ridgecrest-Coso and East Anatolian Fault Zone case, respectively. The solid and dashed lines plot the training and validation accuracy. Different colors denote different model-training data combinations: the blue, cyan, orange, and red lines represent PhaseNet-SCSN/AFAD, PhaseNet-PAL, SAR-SCSN/AFAD, and SAR-PAL, respectively. The accuracy for negative samples is plotted in black lines.

contains a high proportion of false detections, as also reported by Ding et al. (2023). Training data quality also significantly impacts model performance. In the EAFZ case, the PAL training set yields over 60% higher accuracy for both SAR and PhaseNet than the lower-quality AFAD training set (Figure 7b). In Ridgecrest-Coso, this difference is less pronounced due to the higher accuracy of SCSN detections, but PHN-PAL still achieves ~20% higher training accuracy than PHN-SCSN (Figure 7a). Overall, the above results suggest that PAL generates high-quality training sets and SAR demonstrates higher robustness over the PhaseNet model in the training process.

By applying these locally trained phase-picking models in conjunction with PAL associators, we generated six additional catalogs. With the exception of PHN-AFAD, which encountered training issues as mentioned above, all catalogs exhibit consistent FMDs above the magnitude of completeness, indicating no significant missing of larger events, while differing primarily in their lower detection limits (Figures 8c and 8g). All the catalogs also show reasonably low numbers of unassociated picks across stations (Figures 8d and 8h), demonstrating accurate detection performance enabled by local training data. With these catalogs, we conducted two sets of comparisons: (a) SAR-PAL versus PHN-PAL and SAR-SCSN/AFAD versus PHN-SCSN/AFAD, to assess the effect of model architecture; and (b) SAR-SCSN/AFAD versus SAR-PAL and PHN-SCSN/AFAD versus PHN-PAL, to evaluate the impact of training data quality.

4.3.2. Effects of Model Structure

As shown in the model training results, AFAD data is not able to optimize PhaseNet model, which results in nearly no earthquake detections in the EAFZ case (Figures 8e–8g). In contrast, SAR model is more robust to low-quality training data, and SAR-AFAD achieves ~60% of the associated picks compared to SAR-PAL in the EAFZ case (Figure 8f). In other sets of comparisons, PHN-SCSN/PAL can only obtain ~71%–75% number of associated picks and event detections compared to SAR-SCSN/PAL in both cases (Figures 8b–c, f–g). This result aligns with expectations based on the training curve discussed earlier (Figure 7). Comparing to pre-trained PhaseNet, PHN-SCSN/PAL exhibit even lower detection ability in the Ridgecrest-Coso case (Figure 6b); while in the EAFZ case, PHN-PAL only slightly outperforms PhaseNet (Figure 6f). It is noteworthy that PHN-SCSN/PAL models all achieve much higher phase association ratio compared to pre-trained PhaseNet and SAR-PAL, indicating a lower false detection rate.

These results suggest that the training set used for SAR-PAL is not optimized for the PhaseNet model. One possible explanation is that the SAR-PAL workflow samples a higher ratio of negative training data compared to the pre-trained PhaseNet. Additionally, the higher phase association ratio of PHN-SCSN/PAL over SAR-PAL

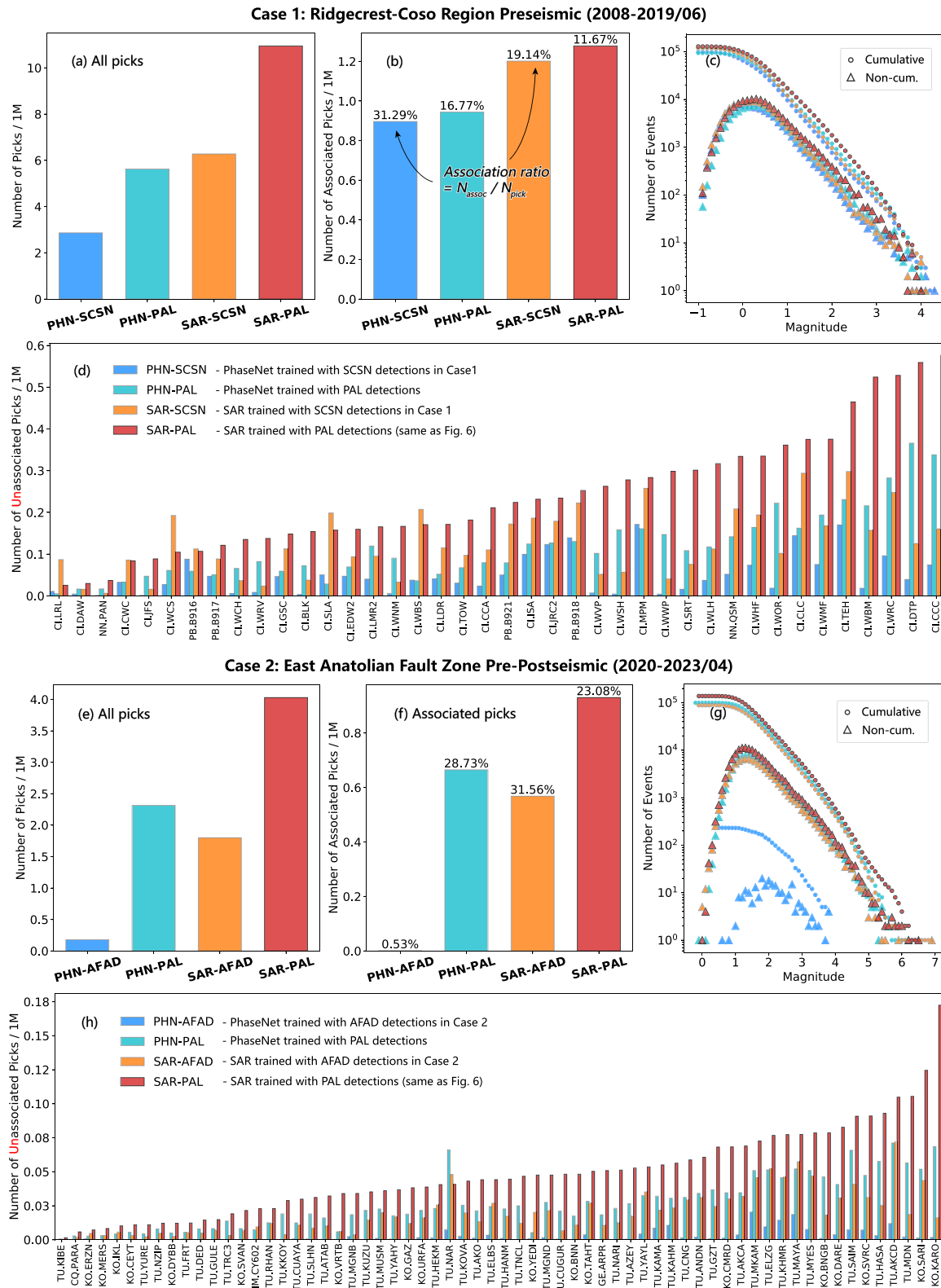


Figure 8. Comparison of phase picking and association performance for locally trained models. The blue, cyan, orange, and red markers denote the results by PhaseNet trained by Southern California Seismic Network (SCSN)/AFAD detections (PHN-SCSN/AFAD), PhaseNet trained by PAL detections (PHN-PAL), Self-Attention RNN (SAR) trained by SCSN/AFAD detections (SAR-SCSN/AFAD), and SAR trained by PAL detections (SAR-PAL, proposed in this study), respectively. All associations are performed using the PAL algorithm. Markers are consistent with those in Figure 6.

may indicate that PHN-SCSN/PAL could be improved by adopting a lower triggering threshold. Thus, while local training data is crucial, the sampling strategy for training data and the triggering threshold must be tailored to specific model architectures. However, identifying the optimal combination of training set and triggering threshold for PhaseNet is beyond the scope of this study.

4.3.3. Effects of Training Data Quality

The differences in training curves analyzed in Section 4.3.1 demonstrate that training sets significantly influence neural network performance. Here, we examine their impact on the resulting earthquake catalogs.

In the Ridgecrest-Coso region, PHN/SAR-SCSN models yielded only a slightly lower number of associated picks compared to PHN/SAR-PAL (Figure 8b), despite achieving an ~ 1.7 -times higher association ratio. This suggests that SCSN constructs a training set of higher accuracy compared to PAL, likely due to manual quality control. Additionally, this result indicates that the phase association process effectively removes a certain level of false detections, implying that a slight reduction in detection accuracy can be acceptable in favor of improved completeness. In the EAFZ case, the differences between PHN/SAR-AFAD and PHN/SAR-PAL are much more pronounced: PHN-AFAD made nearly zero picks, while SAR-AFAD obtained only $\sim 61\%$ of the associated picks compared to SAR-PAL (Figure 8f). As expected, this disparity aligns with the training curves discussed earlier (Figure 7). Interestingly, SAR-AFAD reaches a ~ 1.4 times higher phase association rate over SAR-PAL, suggesting that the SAR model is more conservative when trained on datasets containing a high proportion of false positive labels.

In summary, while using routine catalogs for training may enhance detection accuracy, PAL remains a more reliable training set generator across different regions. Although the detection completeness of PHN/SAR-SCSN could be further improved by lowering the triggering threshold, determining the optimal threshold through trial and error is time-consuming, particularly without an objective criterion for assessing training data quality. Additionally, the high detection accuracy of PHN/SAR-SCSN may partially result from the negative training samples provided by PAL, which are unavailable to users relying solely on the routine catalogs. Finally, it is noteworthy that while the number of picks in the AFAD dataset is comparable to that of PAL (263,437 vs. 286,636), the SCSN dataset contains only $\sim 40\%$ as many picks as PAL (174,405 vs. 451,694). This highlights that the quality, rather than the quantity, of training samples is the critical factor determining AI performance.

4.3.4. Summary

To summarize, our comparisons between locally trained models highlight the importance of adopting an appropriate training data sampling strategy and triggering threshold suited to a specific neural network. While utilizing a local training set improves detection accuracy, ensuring optimal detection completeness requires a high-quality training set, such as the one generated by PAL.

5. Seismicity Imaging With Enhanced Catalogs

Leveraging the high-resolution catalogs developed in Section 3, we conduct practical analysis to demonstrate the advantages of an enhanced catalog in imaging complex fault geometries. Our investigation spans both the Ridgecrest-Coso and EAFZ region, exploring whether the preseismic catalog shed light on coseismic fault behaviors, which is potentially helpful in seismic hazard assessment for other regions prior to large earthquakes.

The seismicity imaging technique relies on the understanding that earthquakes, regardless of their size, occur on faults, including both major and subsidiary ones. Consequently, the clustered seismicity serves as a tool to detect geologically hidden faults, and to directly delineate fault structures at depth with a high resolution (e.g., Hauksson et al., 2012; Ross, Trugman, et al., 2019; Shelly, 2020; Waldhauser & Schaff, 2008). However, the occurrence of seismicity is intricately linked to the stress and frictional state of faults (as reviewed in Bürgmann, 2018), which introduces several caveats in the interpretation of seismicity distribution. For example, certain fault segments can be quiescent during the interseismic period, because of the stress shadow drawn by a deeply penetrated rupture of previous large earthquakes (Jiang & Lapusta, 2016). Therefore, a lack of continuous seismicity does not necessarily imply a discontinuity in fault structure. Another commonly observed phenomenon is the complementary distribution between aftershocks and coseismic slip (Toda & Stein, 2022; Wetzler et al., 2018). In such cases, aftershocks tend to concentrate on the periphery of ruptured asperities, thereby obscure the fault dip in

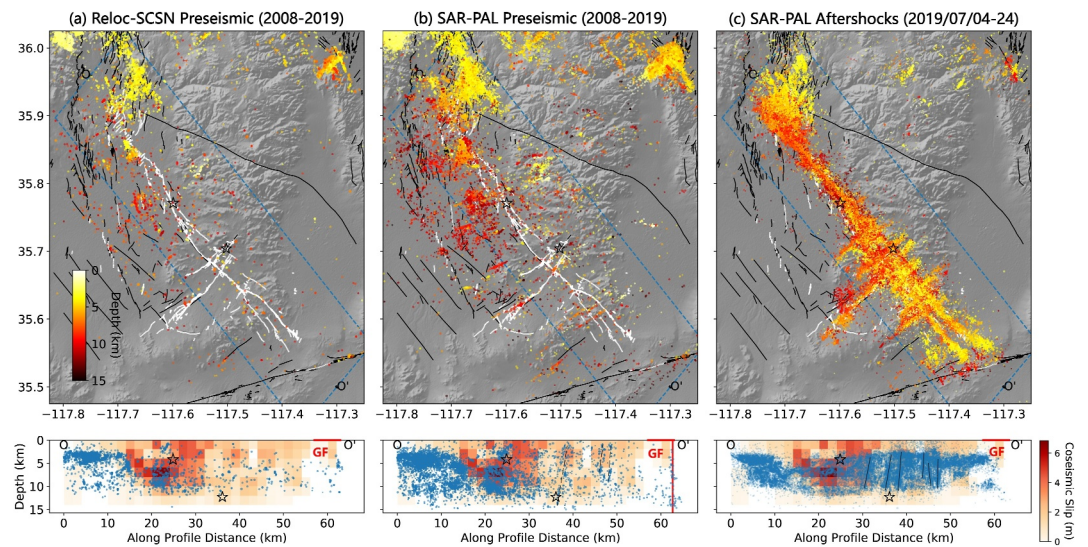


Figure 9. Seismicity on the Ridgecrest faulting area before and after the 2019 earthquakes (a–b) Plot the preseismic period of relocated Southern California Seismic Network catalog (Hauksson et al., 2012) and SAR-PAL catalog in this study, respectively, and (c) for the SAR-PAL catalog aftershock period. Note that the dot size and transparency for the preseismic (a–b) and aftershock (c) period are set differently, because of the very different seismicity rate. In the map view plots (upper panels), the seismicity is plotted as dots that have its color denoting the depth and size varies with the magnitude. The active faults are plotted as black lines; the surface ruptures are marked by white lines. The reference points and spatial coverage of the along-fault cross-section is marked by blue dashed rectangle. The hollow black stars denote the largest foreshock and the mainshock of the 2019 Ridgecrest sequence. In the cross-sections (lower panels), the rupture model by Yue et al. (2021) is plotted as different shades of red. The Garlock Fault on the surface and seismicity-interpreted depth extension is marked by red lines. The black lines are the imaged cross faults by seismicity.

cross-section views. Thus, to accurately infer fault geometry, it is always recommended to jointly consider the focal mechanism solutions and the surface traces of faults or ruptures (e.g., Ding et al., 2023; Lu et al., 2021; Zhou, Ren, et al., 2022).

5.1. 2019 Ridgecrest-Ruptured Faults

As reviewed in Section 3.1.1, the 2019 Ridgecrest sequence activated a largely unmapped orthogonal fault system. However, subsequent detailed geological investigation by Thompson Jobe et al. (2020) suggests that up to 50%–70% of the fault traces could have been mapped before the earthquake. It is practically important to know whether those faults can be imaged by long-term seismicity, and whether this region represents a hazardous seismic gap, considering that the Ridgecrest sequence breaks a ~ 20 year quiescence in the ECSZ (Chen et al., 2020; Hauksson & Jones, 2020; Ross, Idini, et al., 2019).

From the event distribution prior to the earthquake (Figures 9a and 9b), it is evident that an over 20 km-wide zone along the Ridgecrest faults is characterized by spreading microseismicity. This pattern is more distinctly observed in our SAR-PAL catalog compared to both the relocated SCSN catalog and the QTM catalog (Figure 9 and Figure S16 in Supporting Information S1). Though the southern segments have much lower seismicity rate, the seismicity in the SAR-PAL catalog exhibit multiple clusters trending orthogonally to the main faults ruptured during the 2019 M_w 7.1 earthquake, aligning with the focal mechanisms of $M > 2$ events within those clusters (Figure S17 in Supporting Information S1). Additionally, these orthogonal clusters appear to be further activated following the mainshock, since they collocate with the aftershocks and mapped surface traces (Figure 9c). The above observations suggest that multiple subparallel faults extend to the southeast of the 2019 M_w 6.4 foreshock, an area where previously few fault traces had been mapped. The net effect of such a fault system is a distributed shear deformation and a low slip rate on each individual fault, which agree with the weak seismic activity observed in this region.

By examining the seismicity depth distribution in the cross sections (Figure 9), we find a similar pattern of depth contour before and after the 2019 mainshocks, despite contrasting intensities of seismic activity, both

complementing the coseismic slip. Furthermore, the b-value mapping with the SCSN catalog starting from 1980 (Nanjo, 2020) reveals that the areas of weak seismicity in the southern segments correspond to notably low b-values, indicating a high level of differential stress. Collectively, these findings suggest that the southern segments were strongly locked prior to the earthquake, and can be considered as a persistent asperity primed for rupture.

In summary, the distributed microseismicity prior to the mainshock unveils a preexisting fault system composed of multiple subparallel branches. This system, characterized by a weak seismicity and an overall low b-value, represents a persistent asperity of potential hazard.

5.2. Erkenek-Pütürge fault segment (EPF) of EAFZ

As reviewed in Section 3.2.1, the 2023 Turkey doublet is extraordinarily large for continental environments, raising questions about the mechanisms that allow an earthquake to reach such size. Instead of investigating how the rupture grows, we are curious about how it terminates. Specifically, we explore the termination of the 2023 M_w 7.8 rupture at its NE end, named as Erkenek-Pütürge fault segment (EPF), which acted as a barrier also for the 2020 M_w 6.8 earthquake and multiple historical and paleoseismic events (Güvercin et al., 2022; Hubert-Ferrari et al., 2020; Karabacak et al., 2023). Fortunately, the observational condition for this segment of the EAFZ were optimal during our study period from 2020 to 2023, which set a solid foundation to examine the formation of this persistent barrier.

Seismicity from the SAR-PAL catalog clearly reveals an along-strike variation of fault structure complexity (Figures 10a–10c): from a single major fault in profile 01–04, to a major-secondary fault style in profile 05–10, and then to a wide fault zone from profile 11–15, and reverts to the major-secondary pattern in profile 16–18. A zoom-in plot with CC-relocated SAR-PAL catalog shows that the ~10 km wide fault zone (profile 11–15) is composed of two major subparallel branches, along with several subsidiary conjugate faults in between (Figure S23a in Supporting Information S1). This complex inner structure of EPF is also manifested in the highly variable focal mechanisms (Figure S23b in Supporting Information S1). In addition to the complexity variation is a shift in the dip angle of the major fault, which gradually changes from an SSE-dipping in profile 01–04 to an NNW-dipping in profile 17–18 (Figure 10c). Focal mechanism solutions of the 2023 M_w 7.8 mainshock (subevent inversion by Jia, 2023) and the 2020 M_w 6.8 and M_w 5.6 also agree with this contrast in fault dip direction (Figure 10b). The aforementioned structural variations appear to correlate with the coseismic behavior, with the unruptured segments (profile 08–14) show greater geometrical complexity, in contrast to the relatively simpler structures that were ruptured in 2023 (profile 01–07) and 2020 (profile 15–18). However, these detailed features are obscured in the relocated AFAD catalog (Figure S22 in Supporting Information S1, Lomax, 2023). Since fault bends and junctions are known to decelerate or even terminate dynamic rupture (e.g., Duan et al., 2019; Kame et al., 2003; Zhang, Tang, et al., 2023), failing to identify such geometrical complexities may bias the seismic hazard assessments.

The depth distribution of seismicity in the SAR-PAL catalog exhibits a highly concentrated pattern (Figure 10d), aligning with dislocation models that predict stress concentration at the locked-creeping transition depth. Notable changes in seismicity depth distribution are observed at the end of both the 2023 and 2020 mainshocks: at the eastern end of the 2023 rupture, the fault locking depth shallows from approximately 15–8 km; at the western end of the 2020 rupture, a shallow cluster of seismicity is distributed between 0 and 15 km depth, overlapping with the afterslip patch detected via satellite imagery (Cakir et al., 2023). This correlation suggests that fault creep may also contribute to the termination of dynamic ruptures. Notably, there is another ~30 km gap not been analyzed previously, which generates concentrated seismicity at depths of ~8–14 km. Assuming the shallower portion above this concentrated seismicity is locked, this segment probably represents an unruptured asperity that has not experienced any $M > 5.5$ earthquakes to date (till November 2024). Observations of postseismic deformation following the 2023 sequence may help to rule out the alternative hypothesis that the stress in this segment has been relieved through aseismic slip.

In summary, the EPF segment of EAFZ is characterized by complex fault structures, including multiple secondary faults and a conjugate fault system that forms a ~10 km wide fault zone. These geometric complexities, combined with the shallowing of the fault locking depth, potentially render the EPF a persistent barrier to rupture propagation.

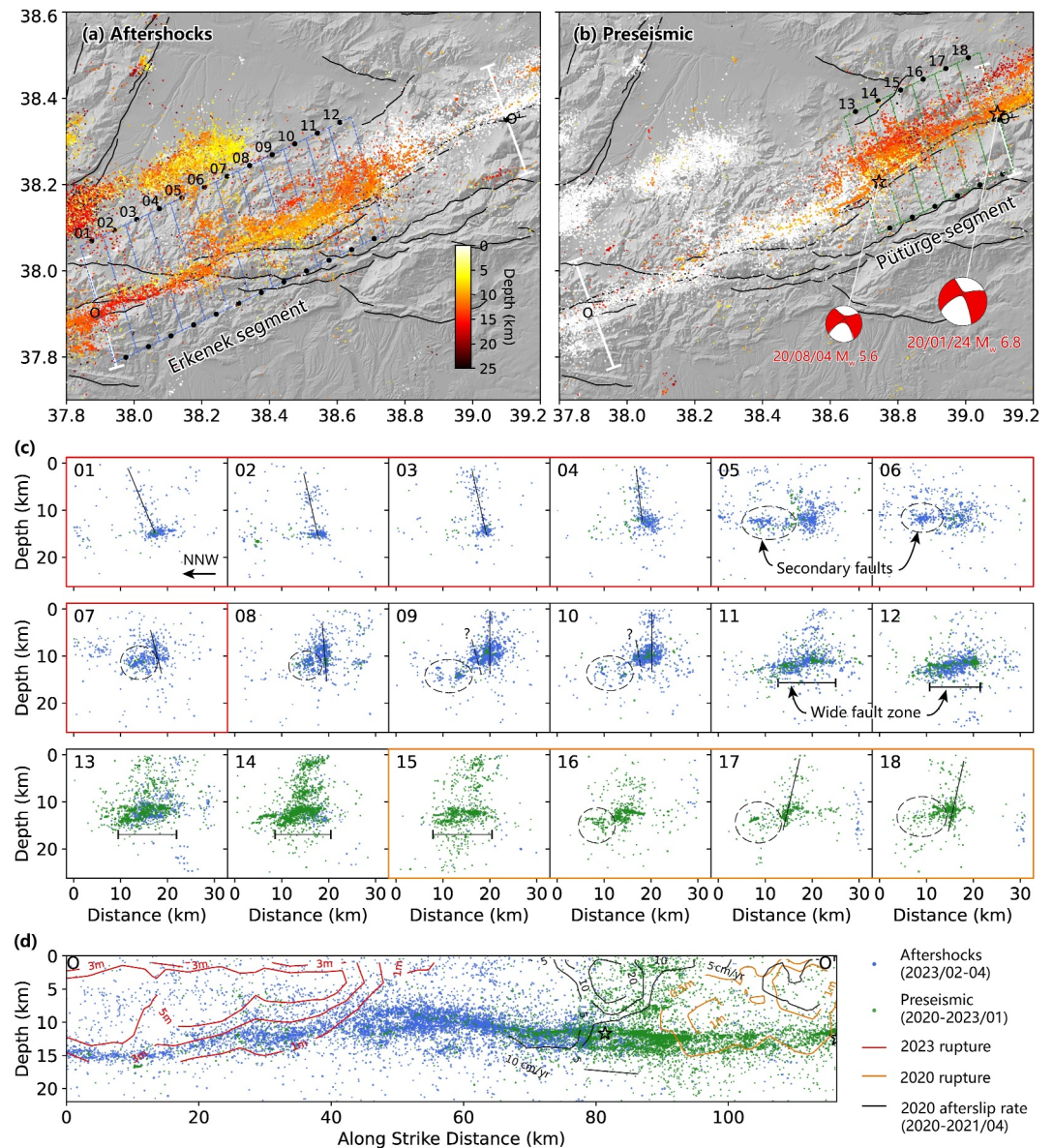


Figure 10. Seismicity resolved by the SAR-PAL catalog and fault geometry interpretation for the Erkenek-Pütürge Fault Segment of East Anatolian Fault Zone (a–b) Plot the map-view distribution of seismicity after (2023/02–04) and before the mainshock (2020–2023/01). The events in current panel/time period are color-coded by the depth, and the seismicity in the other time period is plotted as white dots, (c) Plot the fault-normal cross-sections. The 2023 aftershocks and preseismic events are denoted in blue and green, respectively. The interpreted fault dip is marked as black lines, and (d) Plot the along-strike profile. The coseismic slip of the 2023 M_w 7.8 obtained by Ren et al. (2024), the rupture and afterslip of the 2020 M_w 6.8 by Cakir et al. (2023) are plotted in red, orange, and black contours, respectively.

6. Conclusions

In this study, we introduce AI-PAL, an innovative deep learning-driven workflow for generalized earthquake detection. AI-PAL leverages PAL, an established rule-based workflow, to generate local training data that optimize a SAR model for enhanced phase picking. Through applications to two distinct cases, the Ridgecrest-Coso and EAFZ region, we demonstrated that SAR-PAL realizes a detection ability comparable to MFT, while offering markedly improved temporal stability and computational efficiency. Direct comparisons with other state-of-the-art phase picking and association algorithms demonstrate that SAR-PAL achieves superior scalability, detecting ~ 2.5 times more earthquakes in the EAFZ compared to pre-trained PhaseNet + GaMMA workflow.

Furthermore, we show that even for locally trained models, the performance of neural networks depends not only on the quality of ground truth labels, but also on the alignment between the model architecture and the sampling strategy for training data.

Utilizing the high-resolution catalogs built for the Ridgecrest-Coso and EAFZ regions, we specifically investigated one small fault segment, and imaged much clearer structural features that provide insights into their seismic hazards.

1. The Ridgecrest faults exhibit distributed and weak microseismicity prior to the earthquake, revealing multiple preexisting orthogonal faults. The seismicity depth contour mimics that of the aftershocks and is associated with a low b-value before the mainshock. These observations suggest that the Ridgecrest faults represent a persistent asperity of high stress level.
2. The EPF segment of EAFZ is characterized by complex fault structures, including multiple secondary faults and a conjugate fault system that makes a ~10 km wide fault zone. These geometrical complexities, along with the shallow locking depth and fault creep observed at the terminations of the 2023 and 2020 ruptures, constitute EPF a strong rupture barrier.

Data Availability Statement

Figures in this paper are plotted with GMT and Matplotlib. The continuous seismic data for Ridgecrest-Coso (2008–2019), the waveform-relocated SCSN catalog, the QTM catalog, and the refined focal mechanism solutions by Cheng et al. (2023) are downloaded from the SCSN (SCEDC, 2013). The continuous data for the East Anatolian Fault Zone (EAFZ, 2020–2023/04) is collected through multiple sources: (a) TU network, from the Disaster and Emergency Management Presidency (AFAD, Last accessed 2024/11), downloaded manually, (b) KO network, from Kandilli Observatory And Earthquake Research Institute (KOERI), downloaded with Obspy, and (c) GE, CQ, and IM network, all available through Obspy. The relocated AFAD catalog by Lomax (2023) is available at Zenodo. The focal mechanism solutions for $M_w > 4$ events (2020–2023) are also available at AFAD. The active faults data of the Ridgecrest-Coso region is available at United States Geological Survey (USGS, Last accessed 2024/11); that for East Anatolian Fault Zone comes from the GEM Global Active Faults Database (Styron & Pagani, 2020), and we adopt a more detailed fault data for the EPF area from the Active Faults of Eurasia Database (AFEAD, Zelenin et al., 2022). The surface rupture data of the 2019 Ridgecrest earthquake comes from Ponti et al. (2020); the fault traces of the 2023 Turkey sequence is available by USGS (2024). The coseismic slip model of the 2019 Ridgecrest earthquake comes from Yue et al. (2021); that for the 2023 Turkey earthquake comes from Ren et al. (2024); the rupture model and afterslip model for the 2020 Elazığ (Turkey) earthquake comes from Cakir et al. (2023). The referred velocity models for the RC-Coso region include: CVM-S4 velocity model (Kohler et al., 2003), Hutton et al. (2010) for the whole Southern California, and Shelly (2020) for the Ridgecrest source region. The referred velocity models for the EAFZ include that from Güvercin et al. (2022), Acarel et al. (2019), and Ding et al. (2023). The AI-PAL workflow is open sourced at Github (Zhou, 2024).

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Erratum

The originally published version of this article contained typographical errors. Throughout the article, the section headings were misnumbered. The errors have been corrected, and this may be considered the authoritative version of record.